

PHYS3114 – Electrodynamics

Tutorial 7: Time-dependent potentials and radiation fields

Question 1

- (a) An infinite straight wire carries the current $I(t) = kt$ for $t > 0$ (i.e. current is turned on at $t = 0$). Find the electric and magnetic fields generated at a distance r and time t .
- (b) Repeat for the case of a burst of current $I(t) = I_0\delta(t)$.

[Griffiths 10.9]

Question 2

From the retarded potentials

$$V(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{r} d^3r', \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{r} d^3r' \quad (1)$$

where $t_r = t - r/c$ is the retarded time, derive the fields $\mathbf{E}$ and $\mathbf{B}$.

The derivations can be found in textbooks; these are known as “Jefimenko’s equations” or “Schott’s formulae”.

Question 3

A point electric dipole with moment $\mathbf{p}(t)$ has a fixed position in space. Show that the rate at which energy flows through a spherical surface of radius R centered at the dipole is

$$\frac{dU}{dt} = \frac{2}{3} \frac{1}{4\pi\epsilon_0} \left(\frac{d}{dt} \left[\frac{p^2}{2R^3} + \frac{p\dot{p}}{cR^2} + \frac{\dot{p}^2}{c^2R} \right]_{\text{ret}} + \frac{\ddot{p}_{\text{ret}}^2}{c^3} \right).$$

Question 4

A radio station transmitter consists of a circular current loop acting as a magnetic dipole $\mathbf{m}(t) = I(t)\mathbf{A}$, where $I = I_0 \cos \omega t$ and $\mathbf{A} = \pi b^2 \hat{z}$. The loop is h m above the ground.

- (a) Show that the intensity of radiation is (spherical coordinates)

$$\langle \mathbf{S} \rangle = \frac{\mu_0 m_0^2 \omega^4 \sin^2 \theta}{32\pi^2 c^3 r^2} \hat{r}. \quad (2)$$

The derivation can be found in textbooks.

- (b) Find the intensity of radiation as a function of R , the distance from the base of the tower on the ground (assume a flat earth).
- (c) How far from the base of the tower will the maximum intensity be found?
- (d) If the total power output is 35 kilowatts, its frequency is 90 MHz, the transmitter is $h = 200$ m above the ground, and the transmitter’s radius is 6 cm, determine the maximum intensity.

[Adapted from Griffiths 11.22]

Question 5

Consider an ideal stationary magnetic dipole $\mathbf{m}$ in a static electric field $\mathbf{E}$. Show that the fields carry momentum

$$\mathbf{P}_{\text{EM}} = -\epsilon_0\mu_0(\mathbf{m} \times \mathbf{E}).$$

Hint: Start with the definition $\mathbf{p} = \epsilon_0\mathbf{E} \times \mathbf{B}$, write $\mathbf{E} = -\nabla V$, and use integration by parts to show that

$$\mathbf{p}_{\text{EM}} = \epsilon_0\mu_0 \int V \mathbf{J} d^3r. \quad (3)$$

Equation (3) is valid for any localized static configuration. For a current confined to an infinitesimal neighborhood of the origin we can approximate V using a Taylor series: $V(\mathbf{r}) \sim V(\mathbf{0}) - \mathbf{E}(\mathbf{0}) \cdot \mathbf{r}$. Treat the magnetic dipole as a current loop and make use of the identity $\oint (\mathbf{c} \cdot \mathbf{r}) d\mathbf{l}' = \mathbf{A} \times \mathbf{c}$, where $\mathbf{c}$ is a constant vector and $\mathbf{A}$ is the area of the loop.

Question 6

In lectures we expressed the electric field of a point electric dipole (located at the origin and oriented along the z direction) in cylindrical coordinates as

$$\mathbf{E}(s, z, t) = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{s} \frac{\partial R}{\partial s} \hat{z} - \frac{1}{s} \frac{\partial R}{\partial z} \hat{s} \right) \quad (4)$$

where

$$R(s, z, t) = -s \frac{\partial}{\partial s} \frac{p(t - r/c)}{r}. \quad (5)$$

We showed that curves $R(s, z) = \text{const.} = R_0$ are tangential to the electric field.

The points where the field lines “pinch off” and radiation detaches from the source must occur when $\mathbf{E}(\mathbf{r}, t) = 0$. Show that these points satisfy

$$z = 0 \quad \text{and} \quad \frac{R_0}{s} + \frac{\ddot{p}(t - r/c)}{c^2} = 0.$$